

Lecture 11

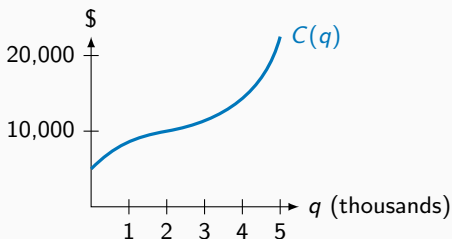
Marginal Cost and Revenue, Focus on Theory: Limits and Derivatives

Big Questions

- Can we relate the derivative to the concepts of marginal cost and revenue from economics?
- Can we find a more concise way of expressing the definition of the derivative?
- What is a limit?

Warm up: The Cost of Producing Pendants

Graphed below is the cost function for producing amethyst pendants.



Poll Question

- (a) Does it cost more to produce the **A** 500th or **B** 2000th pendant?

Note. The cost of producing the 500th pendant (*i.e.* pendant #500), is not the cost of producing the first 500 pendants.

- (b) At roughly which production level q (in thousands) is the marginal cost minimized? **A** 0 **B** 1 **C** 2 **D** 3.5

Profit when Producing Telephones

To produce q telephones, a manufacturer's cost is $C(q)$. To sell q telephones, the revenue is $R(q)$. Suppose that:

$$\begin{array}{ll} C(300) = 5200, & R(300) = 8400, \\ C'(300) = 13, & R'(300) = 8. \end{array}$$

Discussion Question

How much **profit** does the company make by selling 300 telephones?

Estimate the company's profit when producing

$$q = 301, 299, 310, \text{ and } 290 \text{ telephones.}$$

Poll Question

The company is currently producing and selling 300 telephones. In order to generate more profit, they should...

- ☒ **A** increase ☐ **B** decrease ☐ **C** not change

their sales. Assume they sell everything they produce.

Definition

Suppose that cost, revenue and profit of producing q units of a good are respectively:

$$C(q), \quad R(q), \quad \pi(q).$$

- The **marginal cost** (MC) is defined by:

- The **marginal revenue** (MR) is defined by:

- Similarly, the **marginal profit** (MP) is defined by:

- The three quantities are related via:

Velocity of a Football

Suppose that a football kick is filmed. The ball crosses the half-way line 1 second after being kicked, and the distance $d = g(t)$ in meters of the ball from the half-way line is recorded in the table below as a function of time t in seconds.

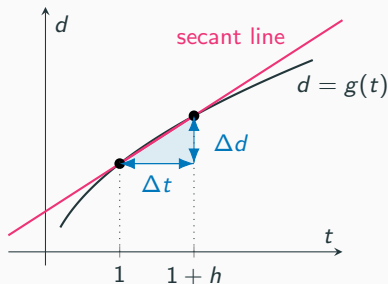
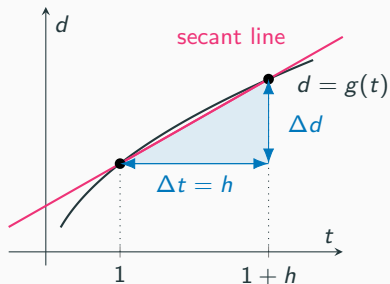
t	1	1.2	1.4	1.6	1.8	2	2.2
d	0.0	6.0	11.8	17.5	22.7	27.5	32.3

Discussion Question

Explain how we can make approximations of the ball's velocity as it crosses the half-way line that successively become more and more accurate.

Visualizing the Approximations

Our method is to take the **limit** of the of g over shorter and shorter intervals with 1 as an end point (i.e. intervals containing 1).

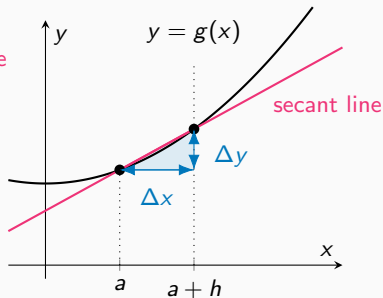
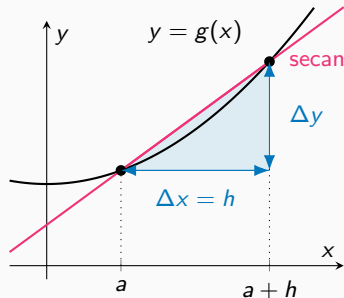


We can express this idea more concisely using *limit notation*:

$$g'(1) =$$

More Generally

Suppose that $y = g(x)$ is any function. Then we can express $g'(a)$ as a **limit** of average rates of change:



Using *limit notation*:

$$g'(a) =$$

What is a limit?

Definition

We say that $f(x)$ **has limit** L as x **approaches** a and write

$$\lim_{x \rightarrow a} f(x) = L$$

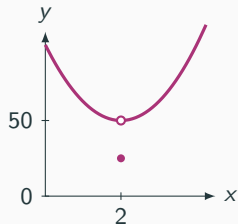
provided that we can make $f(x)$ as close to L as we like, by taking x sufficiently close (but not equal) to a .

If $f(x)$ does not approach a single value as x approaches a , then we say that $\lim_{x \rightarrow a} f(x)$ **doesn't exist**.

Poll Question

The graph of a function $y = f(x)$ is on the right. Appealing to the above definition, does f have a limit as x approaches 2? Explain.

- A** Yes **B** No **C** We can't tell



A Limit

Please set your calculator to radians mode (for instance, 360° becomes 2π).

Discussion Question

What is $L = \lim_{x \rightarrow 0} \frac{\sin(5x)}{x}$ (if it exists)?

To investigate, please use your calculator to fill in the table below.

x	-0.1	-0.01	-0.001	0	0.001	0.01	0.1
$\frac{\sin(5x)}{x}$							

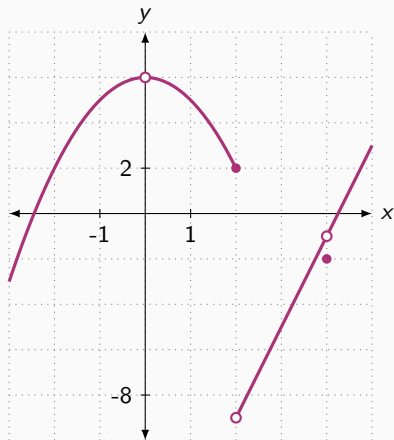
- Based on this data, we expect that $\lim_{x \rightarrow 0} \frac{\sin(5x)}{x} = \boxed{}$ if it exists.

After Class

Investigate the limit using [Desmos](#).

Limits, limits, limits

Below is the graph of $y = g(x)$.



Discussion Question

What are the following limits (if they exist)? Explain using the definition of a limit.

(a) $\lim_{x \rightarrow -2} g(x) =$

(b) $\lim_{x \rightarrow 0} g(x) =$

(c) $\lim_{x \rightarrow 2} g(x) =$

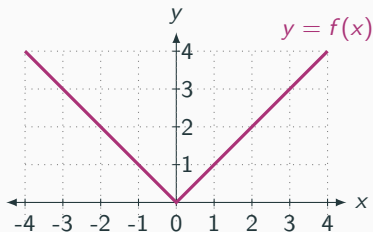
(d) $\lim_{x \rightarrow 4} g(x) =$

Limit Definition of the Derivative

Definition

If the limit exists, **the derivative** $f'(a)$ of a function $y = f(x)$ at a point $x = a$ is the limit:

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}.$$



Discussion Question

Using the limit definition, discuss how to find the following derivatives, if they exist:

(a) $f'(2) =$

(b) $f'(-1) =$

(c) $f'(0) =$

To do before next class:

- Complete Week 3 Checkpoint Workbook!
- Read the Focus on Theory section at the end of Chapter 2.
- Complete Homework 3.
- Week 3 Reflection Journal
- Work on the first draft of your Pod Project Report (due Tuesday, Oct 6).

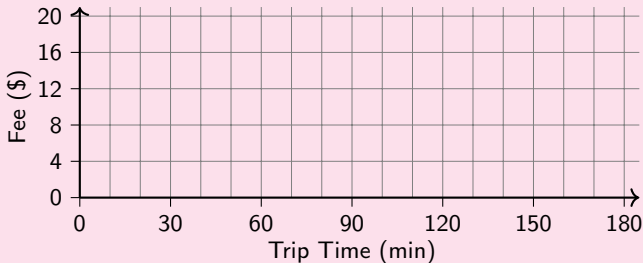
Have a great day!

Motivation for next time: Bike Share Toronto

If you have an annual membership with Bike Share Toronto, bike trips of up to 30 minutes are free. However, if your trip lasts for longer than 30 minutes and up to an hour, a fee of \$4 will be charged. If your trip lasts for longer than 60 minutes and up to 90 minutes, you'll be charged \$8. This pattern continues for each 30 minutes, up to 3 hours.

Discussion Question

Graph the fees as a function f of trip time on the grid below.



What do you notice?

Continuity

Definition

The function f is **continuous** at $x = c$ if f is defined at $x = c$ and

$$\lim_{x \rightarrow c} f(x) = f(c).$$

A function is continuous on an interval (a, b) if it is continuous at every point in the interval.

Discussion Question

The graph of f is to the right.

At what points is f continuous?

